



FOR JAVA DEVELOPERS MOVING TO NESTJS

Test 2: two calls.

Spring Boot to NestJS, episode 17: unit testing with Vitest

Mockito and Vitest make different choices about mock lifecycle and strictness, and the defaults are version-sensitive.

A mock shared by two tests carried its calls into the second test on Vitest 4.1, the version nest new installs at the time of recording. Vitest 5 and Mockito both reported one call each.

clearMocks is the Mockito-like reset. mockReset destroys shared stubs before the first test.

Mockito fails an unused stub; Vitest passes it. A vi.mock factory cannot read a file constant, because the call is hoisted.

In scope

- @Mock and @MockitoBean against vi.fn and overrideProvider
- Mock lifecycle between tests
- clearMocks, mockReset, restoreMocks
- Strict stubs and hoisting

Not in scope

- Fake timers
- Integration tests: next episode

Measured on Vitest 5.0.3 and on the Vitest 4.1.11 a fresh nest new installs.

01 The measurements

MOCKITO, @Mock FIELD	calls seen: 1, then 1
VITEST 4.1 (nest new today), DEFAULT	calls seen: 1, then 2
VITEST 5.0, DEFAULT	calls seen: 1, then 1
clearMocks	history cleared, stub kept
mockReset	describe-scope stub undefined from test 1
restoreMocks	spy back to the real method
UNUSED STUB, MOCKITO	UnnecessaryStubbingException
UNUSED STUB, VITEST	passes, no warning
vi.mock FACTORY READS A FILE CONST	ReferenceError: before initialization

One mock shared by two tests, each calling it once.

02 On the Vitest a new project gets

@nestjs/schematics 12.0.6 pins "vitest": "^4.1.2", which never reaches 5. Set clearMocks, or upgrade to Vitest 5 deliberately after its migration notes.

```
vitest.config.ts

import { defineConfig } from 'vitest/config';

export default defineConfig({
  test: { clearMocks: true },
});
```

shared mock: Vitest 4.1 default 1 then 2; with clearMocks 1 then 1; Vitest 5.0 default 1 then 1

Run on a real machine, 2026-09-30, by scripts/verify-testing.sh and verify-testing-v4.sh in the course repository.

03 The container-level swap

@MockitoBean replaces a bean in the Spring test context; overrideProvider replaces a provider in the Nest testing module. At the time of recording, @MockBean is not in Spring Boot 4.

```
override.spec.ts

const moduleRef = await Test.createTestingModule({ controllers: [AppController], providers:
[AppService] })
  .overrideProvider(AppService)
  .useValue({ getHello: vi.fn().mockReturnValue('stubbed') })
  .compile();
```

overrideProvider: the controller returned "stubbed"; @MockitoBean: the injected bean returned "stubbed"

Run on a real machine, 2026-09-30, by scripts/verify-testing.sh and verify-testing-v4.sh in the course repository.



Before you trust a Vitest suite like a Mockito one

1. Check which Vitest major your project is on
2. On Vitest 4, set clearMocks: true
3. Do not turn on mockReset with describe-scope stubs
4. Create shared stubs in beforeEach
5. Keep vi.mock factories self-contained
6. In NestJS, prefer overrideProvider over module mocks

Everything else

 Site https://code-with-sam-dev.github.io	 Facebook https://www.facebook.com/codewithsamdev
 YouTube https://www.youtube.com/@CodewithSam-Dev	 GitHub https://github.com/code-with-sam-dev
 LinkedIn https://www.linkedin.com/company/code-with-sam-dev	 Buy me a coffee https://buymeacoffee.com/codewithsam
 x https://x.com/CodeWithSamDev	 Buy me sadza https://buymesadza.com/codewithsam
 TikTok https://www.tiktok.com/@codewithsamdev	Every measurement, one command: scripts/verify-testing.sh https://github.com/code-with-sam-dev/spring-to-node
 Instagram https://www.instagram.com/codewithsamdev	Vitest migration guide https://vitest.dev/guide/migration.html

If this was useful, a like and a subscribe help more than you would think. And tell me in the comments which part of your workflow you want taken apart next.

Spring is a trademark of Broadcom. Java is a trademark of Oracle. Node.js is a trademark of the OpenJS Foundation. This is an independent, unofficial guide produced by Code with Sam, not affiliated with or endorsed by any of them.

Technical claims verified 2026-09-30 against the documentation named beside each one. Vendors change behaviour and defaults between versions, so check the version you are actually running before you rely on any of it.